ASSIGNMENT - I

PREDICTIVE MAINTENANCE ON SENSOR DATA

Analyze time-series data from industrial sensors to predict when a machine is likely to fail, using classical and ML-based approaches.

Tasks

Data Requirements:

- Use provided datasets (e.g., temperature, vibration, voltage) from simulated machinery
- Clean and preprocess time-series data (missing values, resampling, smoothing)
- Feature engineering (e.g., moving averages, standard deviation, trend detection)
- Build baseline model using classical methods (ARIMA or logistic regression)
- Build ML model (Random Forest or XGBoost) for predicting failures
- Evaluate models using accuracy, F1-score, and confusion matrix

Deliverables:

- Python code (cleaned notebook or script)
- A short report comparing the two models and interpreting feature importance
- Confusion matrix, graphs of predicted vs actual

Submit:

Python script + PDF Report (3–5 pages)

Estimated time: 2 weeks